

PAPER_1553 — Hubble Constant $H_0 = H_0 = K_{\text{MEX}} \cdot D_{\text{crit}} + (D + SO_5) - 2 \cdot F_{\text{TRZ}} \cdot D + F_{\text{TRZ}}^2 \cdot D + F_{\text{TRZ}}^2 \cdot SSQ^2 \approx 67.41 \text{ km/s/Mpc}$

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** Cosmology **Date:** June 18, 2026 **Status:** CLOSED — 0.015% closure from PAPER_1209GG_S648

Observation

PAPER_1209GG_S648 (Cosmology Unified Proof Set) lists **Hubble Constant H_0** with the integer-primitive expression:

Hubble Constant $H_0 = H_0 = K_{\text{MEX}} \cdot D_{\text{crit}} + (D + SO_5) - 2 \cdot F_{\text{TRZ}} \cdot D + F_{\text{TRZ}}^2 \cdot D + F_{\text{TRZ}}^2 \cdot SSQ^2 \approx 67.41 \text{ km/s/Mpc}$
Status: **0.015%**.

UQFF Closed Identity

$H_0 = K_{\text{MEX}} \cdot D_{\text{crit}} + (D + SO_5) - 2 \cdot F_{\text{TRZ}} \cdot D + F_{\text{TRZ}}^2 \cdot D + F_{\text{TRZ}}^2 \cdot SSQ^2 \approx 67.41 \text{ km/s/Mpc} \quad 0.015\%$

Physical Interpretation

$H_0 \approx K_{\text{MEX}} \cdot D_{\text{crit}} = (25/12) \cdot 26 \approx 54.17 + (D + SO_5) = +14 \rightarrow 68.17$ with small corrections to land at 67.41. Matches Planck 2018 (67.4 km/s/Mpc) to 0.015%.

NOT REPLACEMENT

CODATA/PDG treats Hubble Constant H_0 as a precision-measured constant. UQFF supplies the structural integer-primitive form, demonstrating the framework's predictive economy without claiming to replace experimental measurement.

Reference

- Source: PAPER_1209GG_S648
- Related: PAPER_1209 series unified proof sets
- Calculator dispatch: `calculate_paradox({"paradox": "h0_planck_67_41"})`

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